

BEGISHEVO, Russian FederationWMO: **276070**Lat: **55.565N**Lon: **52.093E**Elev: **196**StdP: **98.99**Time Zone: **3.00 (E03)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.8	-24.9	-30.2	0.2	-26.9	-27.8	0.3	-24.9	14.2	-4.2	12.8	-5.8	3.4	230	0.726

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	31.2	19.5	29.1	19.1	27.1	18.3	21.1	27.7	20.2	26.9	19.5	25.8	4.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.1	23.8	18.1	13.3	22.5	17.1	12.5	22.3	61.9	28.5	59.0	26.7	56.3	25.8	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.7	9.4	-29.9	33.4	3.3	1.9	-32.2	34.8	-34.1	36.0	-36.0	37.1	-38.4	38.5		
(5)			-29.9	22.9	3.2	1.8	-32.2	24.2	-34.0	25.3	-35.8	26.3	-38.1	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-12.4	-11.6	-4.6	5.1	13.8	17.4	19.5	18.3	11.9	4.4	-2.5	-8.6
	DBStd	12.60	7.17	6.50	4.87	4.96	4.99	4.60	3.90	4.52	4.13	4.63	5.34	6.76
	HDD10.0	3098	693	604	453	163	21	4	0	1	26	183	374	578
	HDD18.3	5302	952	838	711	398	155	71	32	57	196	433	623	836
	CDD10.0	1026	0	0	0	15	139	226	296	258	84	8	0	0
	CDD18.3	188	0	0	0	0	16	42	69	57	4	0	0	0
	CDH23.3	1795	0	0	0	3	182	385	646	549	31	0	0	0
	CDH26.7	567	0	0	0	0	35	111	211	207	3	0	0	0
Wind	WSAvg	5.1	5.3	5.0	5.5	5.4	5.1	4.7	4.3	4.3	4.6	5.5	5.7	5.4
Precipitation	PrecAvg	537	36	27	30	28	45	58	54	64	53	53	46	42
	PrecMax	720	77	54	64	62	110	137	114	166	118	114	77	111
	PrecMin	284	10	4	3	0	8	6	10	7	5	14	2	5
	PrecStd	92	15	14	17	19	28	29	25	35	30	25	22	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.8	1.9	7.9	23.0	29.9	32.2	34.0	35.0	26.2	18.8	10.9	2.8
		MCWB	1.3	0.6	3.8	13.3	17.4	19.3	19.8	19.9	15.5	10.6	9.5	1.8
	2%	DB	0.2	0.9	5.0	19.1	27.2	29.9	31.0	31.9	23.8	15.2	8.0	1.8
		MCWB	-0.1	0.2	2.0	11.2	16.2	19.4	20.0	19.5	14.9	9.3	7.3	1.3
	5%	DB	-1.0	-0.8	3.0	16.2	25.2	27.2	29.0	28.8	20.9	13.0	5.9	0.9
		MCWB	-1.3	-1.4	1.0	9.2	15.3	18.6	19.4	19.1	14.4	9.1	5.0	0.6
	10%	DB	-2.9	-3.0	1.8	13.1	23.0	25.2	26.9	26.1	18.8	11.0	3.9	-0.1
		MCWB	-3.4	-3.8	0.4	7.7	14.1	17.6	18.8	18.2	13.3	8.0	3.2	-0.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	0.8	4.1	13.7	19.2	21.8	22.8	21.3	17.5	12.2	9.7	2.1
		MCDB	1.4	1.1	7.0	21.5	27.5	28.8	28.9	28.2	22.9	15.8	10.6	2.1
	2%	WB	-0.1	0.0	2.4	12.1	17.5	20.8	21.4	20.6	16.1	10.6	7.1	1.1
		MCDB	0.1	0.5	4.2	18.2	25.4	27.6	27.2	28.2	20.9	13.6	7.8	1.3
	5%	WB	-1.5	-1.5	1.3	10.0	16.2	19.7	20.3	19.8	14.9	9.6	4.9	0.3
		MCDB	-1.2	-0.8	2.4	14.9	23.2	25.4	26.4	27.4	19.2	11.9	5.5	0.6
	10%	WB	-3.3	-3.8	0.5	8.1	14.9	18.5	19.5	19.0	13.9	8.5	2.9	-0.7
		MCDB	-2.9	-3.2	1.4	12.5	21.3	23.8	25.5	25.1	17.9	10.9	3.3	-0.4
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.4	7.0	8.4	11.0	10.1	10.0	10.0	8.3	6.0	4.2	4.9
		MCDBR	4.7	5.5	6.9	13.1	13.5	12.5	12.8	13.3	11.8	9.5	4.9	2.9
		MCWBR	4.9	5.3	5.2	7.2	6.2	5.9	5.5	5.1	5.9	6.0	4.4	3.0
	5% WB	MCDBR	4.7	5.3	5.9	12.0	12.6	11.4	10.7	11.9	10.5	8.1	4.8	3.0
		MCWBR	4.8	5.3	4.8	7.1	6.4	6.1	5.4	5.3	5.8	6.0	4.6	3.2
Clear-Sky Solar Irradiance	taub		0.291	0.315	0.365	0.406	0.391	0.385	0.404	0.428	0.375	0.350	0.313	0.288
	taud		2.042	2.040	1.976	2.110	2.298	2.359	2.309	2.221	2.348	2.304	2.178	2.025
	Ebn at Noon		584	725	774	807	850	860	833	780	778	694	565	482
	Edh at Noon		67	98	134	136	120	114	118	121	93	76	59	54
All-Sky Solar Radiation	RadAvg		0.74	1.74	3.13	4.42	5.45	5.93	5.82	4.53	2.82	1.29	0.54	0.41
	RadStd		0.10	0.19	0.34	0.38	0.49	0.54	0.52	0.39	0.34	0.26	0.10	0.10